Docking-Based Resistance-Aware Prioritization of Antimalarial Leads from African Natural Products with Targeted Molecular-Dynamics Follow-up

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Abstract

Resistance-aware docking may help prioritize antimalarial leads for follow-up, but docking scores do not establish target engagement or mechanism of action. We evaluated 17 candidates across 136 PfDHFR/PfCRT wild-type and mutant docking systems, defining a target-balanced primary set of 12 candidates with eligible wild-type docking scores for both targets and a coverage-limited sensitivity set of 5 PfCRT-only candidates. Under a mutually exclusive docking-score-retention scheme, the primary set

contained 1 Class A*, 1 Class A, 4 Class B, 5 Class C, and 1 Class D candidate; the available-target sensitivity analysis yielded 5 Class A*, 1 Class A, 5 Class B, 5 Class C, and 1 Class D. In the target-balanced set, the PNS–RRS association was weak and uncertain ($\rho = -0.2098$, two-sided permutation $p = 0.5144$, Bonferroni-adjusted $p = 1.0000$, $n = 12$); the nominal negative association in all 17 candidates ($\rho = -0.5588$, permutation $p = 0.0222$, adjusted $p = 0.0667$) is reported only as a coverage-sensitive exploratory result. Two of 17 candidates (11.8 %) had ACSI > 0.70, and cohort-scoped weight perturbations gave Spearman correlations of 0.9167–0.9804. In a separate parent-study MD cohort, two of four complexes retained their ligands under the stated geometric criterion, but only PfCRT–214 yielded an interpretable endpoint estimate ($(-18.25 \pm 0.40)\,\mathrm{kcal\,mol^{-1}}$). A separate 16-system, single-replicate Set-C MD pilot did not reproduce the docking-based retention pattern as a trajectory-derived energetic signal. PfCRT results remain conditional on the neutral-pH protonation protocol and should not be extrapolated to the acidic digestive-vacuole environment. Biological polypharmacology, resistance circumvention, and pathway-level mechanism were not tested.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden[1]. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries, and declining efficacy of some artemisinin partner drugs has been reported[1,2]; spatial-temporal modelling of *pfk13* markers further indicates that the geographic footprint of these mutations in Africa is expanding[3]. Genomic surveillance further documents the co-occurrence of *Pfcrt* and *Pfdhfr* resistance variants across diverse transmission settings[4], underscoring the relevance of multi-target resistance panels. Resistance to partner drugs in artemisinin-based combination therapies, driven by mutations in *Pfcrt*, *Pfdhfr*, and *Pfdhps*, further narrows the therapeutic options available to clinicians[5,6].

Polypharmacology is increasingly treated as a design strategy for addressing complex disease biology, and multi-target-directed ligands represent an important translational class in contemporary pharmacology[7]. The rationale is evolutionary as well as pharmacological: if a compound requires simultaneous loss of activity at several relevant targets, the mutational barrier to treatment failure may increase. That proposition is conditional, however. A list of predicted protein contacts is not a mechanism-of-action assignment; causal network analyses require an explicit perturbation model, a defined biological background, and evidence that the predicted interactions propagate to a phenotype[8]. Recent network-learning approaches to off-target prediction illustrate the value of graph context while also underscoring that network inference remains distinct from experimental target engagement[9]. Natural products are attractive starting points because their stereochemical and scaffold complexity can support activity across multiple protein classes[10,11]. African natural products provide a particularly relevant chemical-space context, with resources such as ANPDB enabling explicit comparison with approved-drug and synthetic reference spaces[12].

In a previous study, we used generative molecular methods and dual-filter consensus docking (AutoDock Vina[13] and DiffDock-L[14]) to generate a library of 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs[15]. We selected the present candidates against a panel spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and the 4GM2 entry represents PfClpR, not PfClpP, and is retained as a proteostasis-related parent-study system. This panel is deliberately a mechanistic cross-section rather than an exhaustive parasite targetome. It allows us to ask whether resistance-aware prioritization remains informative across distinct target classes, while avoiding the stronger claim that the selected compounds define a complete parasite-wide polypharmacology profile.

Rigid docking is efficient for ranking but treats the receptor and scoring landscape in a simplified manner. MD can test whether a docked pose remains compatible with protein flexibility, whereas endpoint methods such as MM-GBSA provide system-specific energetic summaries rather than calibrated thermodynamic observables[16,17]. This distinction is especially important for resistance analysis: a docking RRS is a hypothesis about relative mutant sensitivity, not a measurement of a mutant free-energy difference. Network pharmacology provides a complementary prioritization layer, but centrality is a topological weighting heuristic rather than a causal measure of target essentiality, consistent with the growing interest in multi-target antimalarial design more broadly[18]. We therefore treat these quantities separately: docking-derived RRS for mutant sensitivity, PNS for network-weighted target ranking, ACSI for chemical-space positioning, and targeted MD for post-docking structural scrutiny.

We asked whether target-level docking-score retention could distinguish predicted potency from predicted mutant sensitivity in a chemically diverse antimalarial library. The primary analysis used 12 candidates with eligible PfDHFR and PfCRT wild-type scores; five PfCRT-only candidates formed a coverage-limited sensitivity set. RRS was calculated from empirical Vina scores, whereas PNS and ACSI provided complementary ranking dimensions. Four separate wild-type complexes underwent 10 ns simulations, and a 16-system Set-C pilot provided a secondary test of the docking-based ranking. Biological polypharmacology and resistance circumvention were not tested.

Methods

Study design and estimand separation

The study prespecified one primary estimand and three secondary analyses. The primary estimand was docking-score retention (RRS) in 12 Set-C candidates with eligible PfDHFR and PfCRT wild-type denominators. Secondary analyses comprised the five PfCRT-only candidates, the four-complex parent-study MD check, and the 16-system Set-C pilot. These analyses were kept separate because they address different target coverage, sampling, and measurement questions.

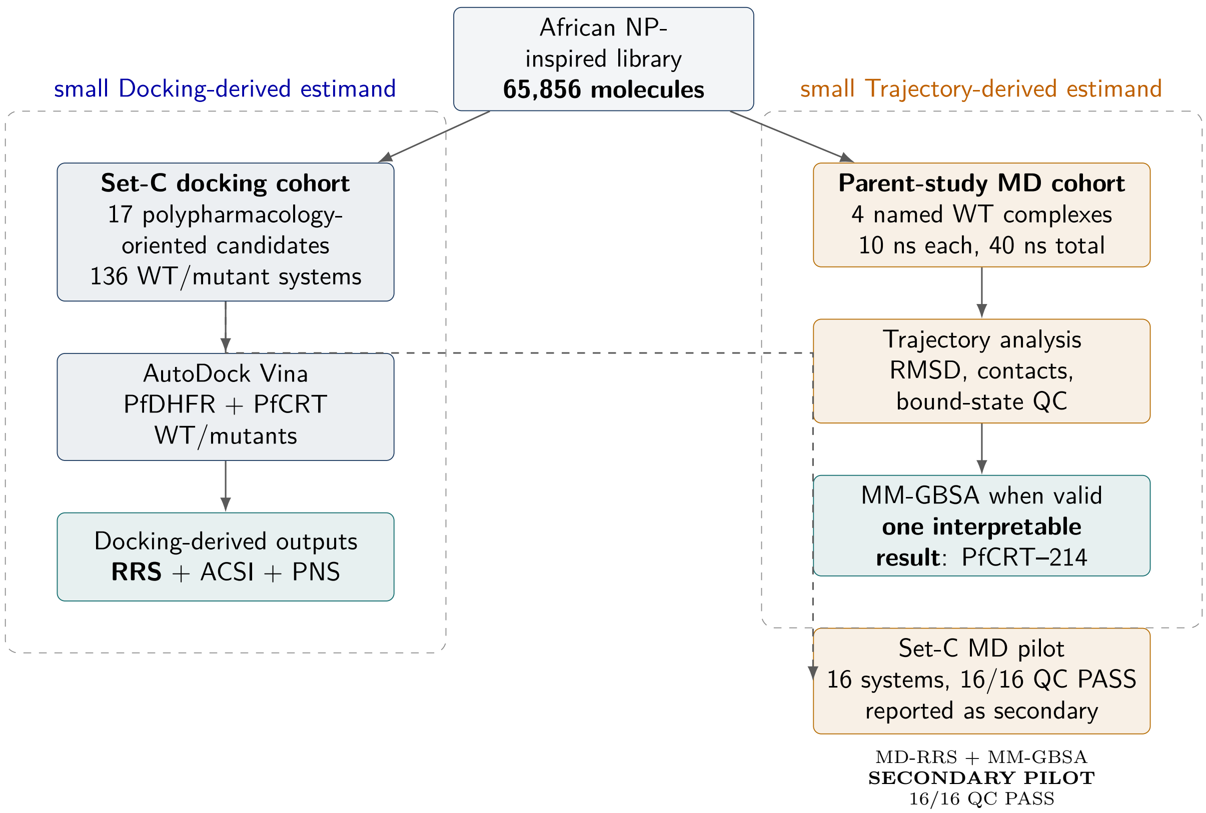

Figure 1: *Study design and separation of computational estimands. The 17-member Set-C cohort supports docking-derived RRS, ACSI, and PNS analyses, whereas the four parent-study complexes support a separate targeted MD and MM-GBSA check. The 16-system Set-C MD pilot (PP-01 and PP-02) is presented separately and is not used to define the docking-RRS classes.*

Candidate selection and filtering

The Set-C cohort is a filtered prioritization cohort rather than a representative sample of the parent library. Class frequencies are therefore conditional on the stated MPO, SYBA, selectivity, multi-target, and scaffold-diversity filters and are not estimates for the full 65 856-molecule library.

Seventeen candidates were selected from the Set-C candidate panel generated from the library of 65 856 molecules described previously[15]. The candidate-selection panel is distinct from the production-MD cohort. Three sequential filters were applied: (1) MPO score $\geq$ 0.70 (primary leads); (2) SYBA score $>$ 0 (synthesisable); (3) selectivity index SI $>$ 10 (selectively antiparasitic). The selection additionally required multi-target engagement of at least two of the four docking targets and scaffold diversity of no more than three compounds per Bemis–Murcko scaffold. The selection includes one of the previously named consensus leads (Ligand 87, PfDHFR). The four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were not part of the 17-candidate docking set but were used to construct the wild-type complexes for the production MD check. The 164 system is identified as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not structural validation of PfClpP. Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were included as reference controls for the docking analysis (Table 1).

Table 1: *Breakdown of the 136 docking protein–ligand systems by category (17 polypharmacological candidates × 8 wild-type/mutant PfDHFR and PfCRT structures).* **Production MD was performed at* 10 ns *each* (40 ns *total) on 4 wild-type parent-study complexes. Ligand 164 is retained as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not presented as PfClpP structural validation. The other systems were Ligand 201 (PfDHFR), Ligand 214 (PfCRT), and Ligand 438 (PfATP4); all were distinct from the 17-candidate docking set.*

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (set C)	17	5	85	—
PfCRT WT + 2 mutants (set C)	17	3	51	—
Docking total	–	–	**136**	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six resistance-associated states for comparison (Table 2); target identifiers are listed in the Supporting Information (Table S5).

Table 2: Resistance mutation panel.

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	PfCRT resistance/comparator state	Not a prevalence estimate

Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation, Ramachandran analysis (MolProbity), and Cα RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0, GMQE exceeded 0.7, Ramachandran favored residues exceeded 95 %, and backbone RMSD was below 1.5 Å. The retained models served as receptor models for docking. Their quality scores do not establish experimental accuracy, and the absence of experimental structures or replicate model ensembles leaves structural uncertainty in the docking results.

Docking protocol assessment

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study[15]: AutoDock Vina 1.2.7[13] with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ[19]), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP).

As an independent boundary audit of these fixed boxes, P2Rank[20] pocket predictions were compared with the grid centers; the concordant PfCRT result and the frame caveats for the other receptors are presented in the Supporting Information (Figure S4). Redocking was considered accurate when RMSD was < 2.0 Å relative to the crystal pose. Results are given separately for the five alignable ligands and for all attempts, including MTX. External enrichment and redocking benchmarks, including the negative DEKOIS 2.0 result, are given in the Supporting Information (Table S1). AutoDock Vina was used for the reported scores; GNINA yielded no usable score.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO–affinity association was null and exploratory; complete correlations are provided in the Supporting Information (Tables S8 and S9).

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information (Figure S1 and Table S3). These predictions were not experimentally validated.

Docking and resistance resilience score

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. GNINA did not yield a score for this panel. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files.

For target t, let $s_{\mathrm{Vina},i,\mathrm{WT},t}$ and $s_{\mathrm{Vina},i,m,t}$ denote the signed Vina scores. Candidates with

__Table 3:__ Target-specific docking-score retention for the 17 *Set-C candidates. RRS is computed separately for each target and only when the corresponding wild-type Vina score magnitude is at least* 5.0 kcal mol^{-1}*. A dash denotes an ineligible target, not a zero score. The primary class applies only to the* 12 *complete two-target candidates; the available-target class is shown for all* 17*. Classes are mutually exclusive: A* requires all available mutant RRS values* $\geq 80\,\%$ *and the minimum eligible wild-type score magnitude* ≥ 7.0 kcal mol^{-1}*; A has the same RRS criterion but a weaker minimum wild-type score; B requires all values* $\geq 70\,\%$ *but not all* $\geq 80\,\%$*; C requires at least one value* $\geq 80\,\%$ *but not all values* $\geq 70\,\%$*; D is the residual class in which no available mutant reaches* 80 %*. All scores are empirical AutoDock Vina scores; RRS is not a free-energy or resistance measurement.*

Candidate	WT PfDHFR	WT PfCRT	Eligible targets	N51I	C59R	S108N	I164L	K76T	K76A	Primary class	Available class
PP-01	7.5	9.3	both	86.4	85.5	87.6	83.9	92.5	90.9	A*	A*
PP-02	–	9.1	PfCRT	–	–	–	–	87.9	94.6	–	A*
PP-03	10.3	11.3	both	68.1	68.3	65.8	69.0	80.0	81.4	C	C
PP-04	8.0	7.0	both	68.8	66.9	68.8	73.5	105.3	103.6	C	C
PP-05	–	10.5	PfCRT	–	–	–	–	77.1	79.3	–	B
PP-06	–	9.9	PfCRT	–	–	–	–	90.6	94.1	–	A*
PP-07	7.6	9.1	both	71.8	71.8	72.4	72.2	84.2	77.6	B	B
PP-08	7.6	7.4	both	62.1	62.1	67.2	66.3	83.9	83.1	C	C
PP-09	7.1	8.6	both	70.6	73.0	75.8	76.5	75.8	73.5	B	B
PP-10	7.7	8.4	both	76.0	76.4	75.7	75.8	89.4	89.6	B	B
PP-11	–	10.0	PfCRT	–	–	–	–	82.7	82.5	–	A*
PP-12	7.5	7.8	both	75.2	66.1	62.9	72.5	70.4	87.2	C	C
PP-13	–	8.3	PfCRT	–	–	–	–	110.2	99.4	–	A*
PP-14	7.6	7.5	both	62.2	65.5	68.0	64.9	78.8	77.7	D	D
PP-15	5.6	9.3	both	123.2	112.7	125.9	131.8	86.2	90.1	A	A
PP-16	7.6	8.7	both	72.9	74.7	73.7	75.9	80.1	80.6	B	B
PP-17	8.3	7.4	both	59.3	61.2	59.5	58.9	76.8	93.4	C	C

$|s_{\mathrm{Vina},i,\mathrm{WT},t}| < 5.0\,\mathrm{kcal\,mol^{-1}}$ are excluded from that target's score-retention analysis. The target-specific Resistance Resilience Score (RRS) is

$$\mathrm{RRS}_{i,m,t} = \frac{|s_{\mathrm{Vina},i,m,t}|}{|s_{\mathrm{Vina},i,\mathrm{WT},t}|} \times 100. \tag{1}$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or biochemical resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. In the eligible candidates, mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); the corresponding target-specific summaries are given in the Supporting Information (Table S13). These values describe differences in score retention across the two target panels, not differential biological resistance. To prevent unequal target coverage from driving the primary inference, the primary target-balanced analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are retained as a coverage-limited sensitivity analysis.

The mutually exclusive classes are defined on the available mutant values, with the target-balanced subset used for the primary class counts:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least 7.0 kcal mol^{-1}.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below 7.0 kcal mol^{-1}.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below 80 %.
- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D**: no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. These thresholds define docking-based triage categories and have not been calibrated against biochemical resistance measurements or clinical outcomes. Eligible-target counts and target-specific score-retention values are given for both analyses. A threshold-sensitivity analysis is provided in the Supporting Information (Table S12); it shows how the class counts change when the WT eligibility and retention thresholds are tightened.

Secondary analyses

The primary RRS analysis was complemented by chemical-space positioning (ACSI), network-weighted ranking (PNS), a targeted four-complex MD check, and a 16-system Set-C MD pilot. These analyses were treated as secondary because they use different estimands and, for the MD analyses, different cohorts and sampling designs. Their numerical results are reported after the primary RRS result and interpreted only within their respective scopes.

African chemical space index

ACSI was used as a cohort-relative chemical-space descriptor. Two of 17 candidates exceeded 0.70 (cohort mean, 0.543); local weight-sensitivity results are provided in the Supporting Information (Table S4). These values describe reference-space positioning, not biological activity or resistance.

Polypharmacology network score

PNS was treated as a secondary network-weighted ranking heuristic. In the target-balanced analysis, PNS–RRS was weak and uncertain ($\rho = -0.2098$, adjusted $p = 1.0000$); complete ranking and imputation details are provided in the Supporting Information (Table S7).

MD stability analysis

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m[21]/GAFF2[22] at 310.15 K using GROMACS[23,24]. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis[25] was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

Table 4: *Production MD stability metrics for wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.*

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Two systems, PfCRT and PfATP4, retained their ligands throughout production. PfCRT had a backbone RMSD of 4.47 Å, a ligand RMSD of 2.45 Å, 78.1 contacts, and 23.1 H-bonds (minimum heavy-atom distance 3.19 Å). PfATP4 shows a backbone RMSD of 12.27 Å, a ligand RMSD of 2.13 Å, 178.4 contacts, and 47.7 H-bonds (minimum distance 2.25 Å). The high backbone RMSD indicates substantial structural drift in this model. Because the simulation omitted a membrane and the topology conversion was not valid for MM-GBSA, the apparent ligand persistence should not be interpreted as validated binding thermodynamics. PfCRT ligand 214 anchors to Lys34 and engages in a robust H-bond network (Gln101, Thr30, Gln297) and a hydrophobic core (Leu301, Leu105, Ala33), consistent with a stable pose in this trajectory; neither specificity nor target engagement is established.

The other two systems, the 164-labelled PfClpR cohort and PfDHFR, had stable protein conformations (backbone RMSD 1.31 Å and 3.05 Å, respectively) but the ligands are completely unbound (minimum distances 67.42 Å and 78.21 Å, zero contacts). This indicates either incorrect initial placement or rapid dissociation during equilibration, not a PBC artefact. Consequently, MM-GBSA and RRS analyses for the 164-labelled PfClpR cohort and PfDHFR are not interpretable as affinity estimates and are excluded from the binding-mode discussion below. Thus, protein equilibration alone was insufficient to retain the docked poses.

The detailed PfCRT–214 RMSD trajectory is shown in Figure 2; per-residue flexibility and contact persistence are provided in the Supporting Information. Per-compound RRS profiles across all six mutants are shown in Figure S2.

MM-GBSA endpoint estimates

MM-GBSA endpoint estimates were obtained for the wild-type systems from their production trajectories (Table 5), using the $\mathrm{GB}^{\mathrm{OBC2}}$ model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3[16] CHARMM36-to-AMBER conversion. Only the PfCRT–ligand 214 system yields an interpretable MM-GBSA endpoint estimate: the `gmx_MMPBSA` native converter successfully produced a valid AMBER topology with the ligand remaining bound throughout production, giving a software-reported endpoint estimate of ΔG_{bind} =

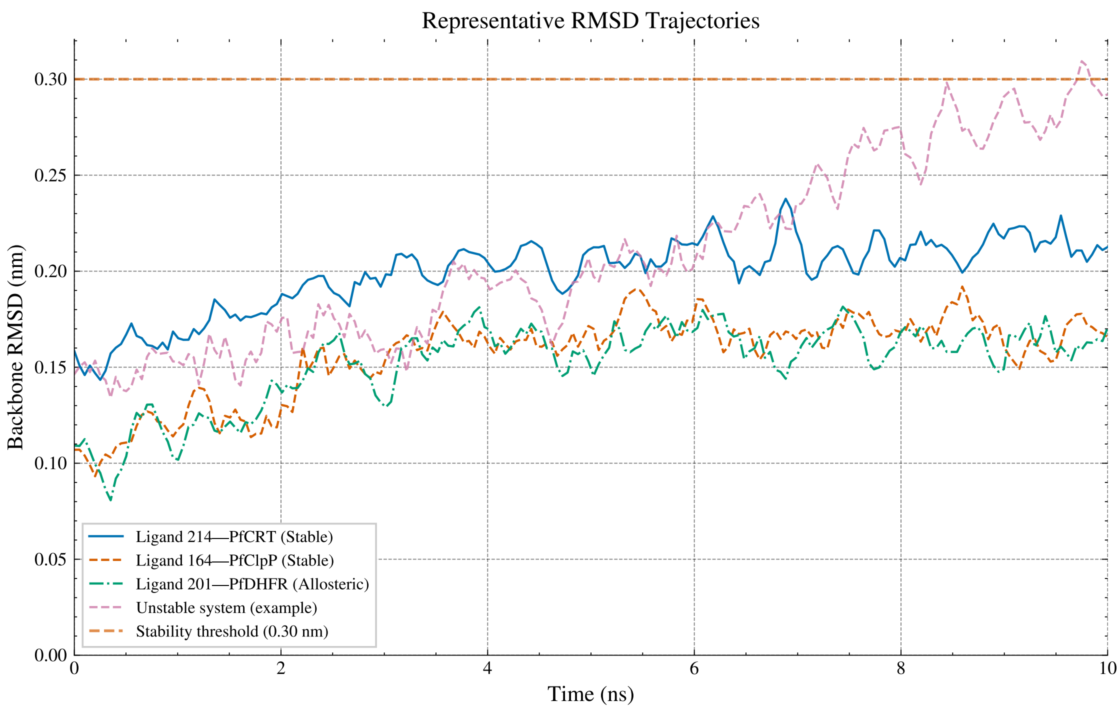

Figure 2: *Detailed stability analysis for the parent-study PfCRT–Ligand 214 complex over* 10 ns *production MD. The system exhibits transporter flexibility while retaining a bound ligand; the primary cohort-level stability values are reported in Table 4.*

(-18.25 ± 0.40) kcal mol^{-1}. The 164-labelled and PfDHFR–201 ligands dissociated during production (minimum distances 67.4 Å and 78.2 Å, zero contacts); their endpoint values are not interpretable for binding-mode analysis. The PfATP4 system was excluded from MM-GBSA because CHARMM36-to-AMBER conversion of its two-chain topology produced a systematic van der Waals artifact (+473 kcal mol^{-1}). Its trajectory supports a bound-state observation in this targeted check, but does not establish converged binding thermodynamics.

All MM-GBSA values reported in this work are endpoint enthalpic scores computed with a single-trajectory protocol; configurational entropy is not included. They are therefore not experimental binding free energies and are interpreted only through within-target contrasts (mutant versus wild type) under identical sampling conditions. Unless stated otherwise, the uncertainty quoted for an endpoint estimate is the standard error of the mean over the 100 stored snapshots; the within-trajectory and propagated standard deviations reported by `gmx_MMPBSA` are tabulated in the supplementary material, and none of these quantities is a replicate uncertainty.

Table 5: *Endpoint interpretation for four parent-study complexes used for production MD. These systems are distinct from the 17 Set-C candidates. Only PfCRT–214 yields an interpretable MM-GBSA endpoint; dissociated systems are reported as N/A.*

Ligand	Target	ΔG_{bind} (kcal mol^{-1})	SEM	Interpretation
214	PfCRT-WT	−18.25	0.40	Interpretable; ligand bound†
164	PfClpR-labelled cohort-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated§
438	PfATP4-WT	–	–	N/A; conversion corrupted*

†Ligand remains bound (minimum distance 3.19 Å, 78 contacts); conversion via `gmx_MMPBSA` native converter.
The quoted uncertainty is the standard error of the mean over the 100 stored snapshots (see Methods, MM-GBSA binding free energies).
‡Ligand dissociates to 67.4 Å minimum distance with zero contacts during production; the reported value is a non-physical end-point estimate for an unbound state and is excluded from binding-mode conclusions.
§Ligand dissociates to 78.2 Å minimum distance with zero contacts; same caveat applies.
*Excluded: conversion artifact in the two-chain PfATP4 topology (+473 kcal mol^{-1} van der Waals inflation); structural persistence is reported separately.

Correlation between methods

The four parent-study MD systems were not members of the 17-candidate Set-C cohort; consequently, no Vina–MM-GBSA correlation is estimated across cohorts. Vina–DiffDock comparisons and the exploratory Ersilia activity-model analysis are reported in the Supporting Information as contextual analyses of complementary ranking dimensions.

Consensus leads

The intersection of high-RRS (Class A or B), high-ACSI (>0.5), and high-PNS compounds defines a computational prioritization set for future experimental testing. These compounds combine resistance-resilience, African-NP-character, and network-weighted docking signals; the intersection is not itself experimental validation.

Cross-metric correlation: PNS, ACSI, RRS, and docking scores

In the target-balanced set, PNS and RRS were weakly associated ($\rho = -0.2098$, permutation $p = 0.5144$, adjusted $p = 1.0000$; bootstrap 95 % interval $[-0.7582, 0.5429]$, $n = 12$). The 17-candidate estimate was more negative ($\rho = -0.5588$, permutation $p = 0.0222$, adjusted $p = 0.0667$; interval $[-0.7891, -0.1349]$), but unequal target coverage limits its interpretation.

ACSI and RRS were also weakly associated ($\rho = -0.4056$, permutation $p = 0.1922$, adjusted $p = 0.5766$; bootstrap interval $[-0.8917, 0.2857]$). The 17-candidate estimate was weaker ($\rho = -0.1324$, $p = 0.6117$). The data do not resolve a relationship between chemical-space position and score retention.

RRS and the weakest eligible WT score were not associated ($\rho = -0.1661$, permutation $p = 0.6038$, adjusted $p = 1.0000$; bootstrap interval $[-0.7300, 0.5932]$, $n = 12$). PNS and WT docking scores are mechanically related because PNS incorporates WT docking scores; this relationship was not treated as an independent hypothesis.

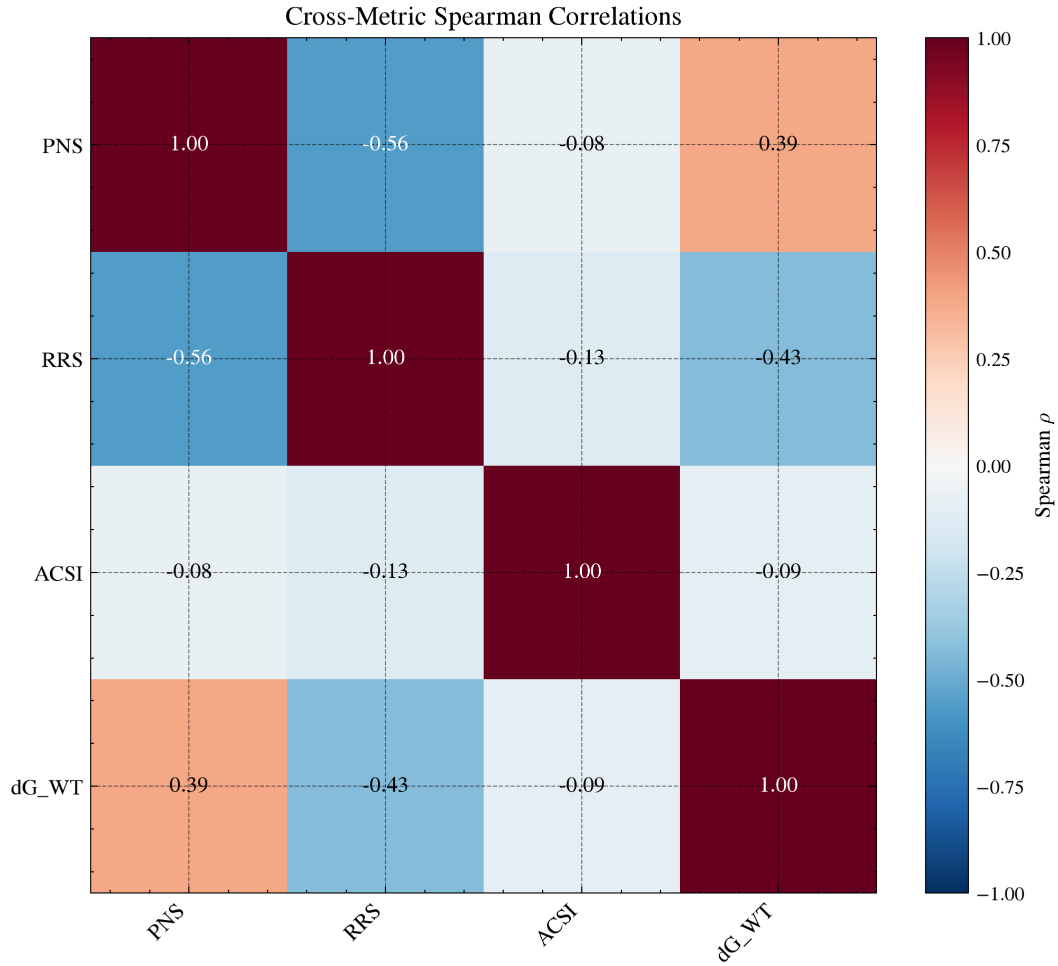

Figure 3: *Cross-metric correlation matrix for the Set-C candidates. The heat map shows Spearman ρ values; the target-balanced primary analysis uses the 12 complete two-target candidates, while the 17-candidate matrix is a coverage-sensitive descriptive view. PNS and WT docking score are mechanically linked and are not treated as an independent hypothesis.*

Table 6: *Cross-metric Spearman correlations. The complete two-target set (n = 12) is primary; the 17-candidate set is a coverage-sensitive secondary analysis. Permutation p-values use* 100 000 *permutations; adjusted p-values apply Bonferroni correction to H1–H3. Bootstrap intervals are percentile* 95 % *intervals from* 10 000 *resamples.*

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95 % interval
PNS vs. RRS	H1	−0.209 8	0.514 4	1.000 0	[−0.7582, 0.5429]
ACSI vs. RRS	H2	−0.405 6	0.192 2	0.576 6	[−0.8917, 0.2857]
RRS vs. weakest WT score	H3	−0.166 1	0.603 8	1.000 0	[−0.7300, 0.5932]

Analysis populations. The table reports the primary 12-candidate complete two-target set. In the coverage-sensitive 17-candidate set, PNS–RRS gave $\rho = -0.5588$ (permutation $p = 0.0222$, adjusted $p = 0.0667$), ACSI–RRS gave $\rho = -0.1324$ ($p = 0.6117$), and weakest-WT RRS gave $\rho = 0.1759$ ($p = 0.4947$).

Set-C MD pilot: trajectory-derived MD-RRS and MM-GBSA

A secondary 16-system Set-C MD pilot was used to test whether trajectory-derived geometry and MM-GBSA endpoints reproduced docking-RRS patterns. All systems passed the prespecified trajectory-QC gate, but each mutant state was represented by a single 10 ns replicate. Across the eight mutant states with comparable docking references, docking predicted weaker mutant scores, whereas trajectory-derived metrics generally indicated retained or tighter local contacts; seven of eight direction comparisons were discordant. MM-GBSA ratios exceeded 100 % for eight of twelve mutant states, but this ratio is an endpoint comparison rather than a free-energy resilience measure. The pilot yielded a different pattern from docking-RRS; the discrepancy does not establish resistance or enhanced binding. Complete values and QC annotations are provided in the Supporting Information (Figure S3 and Table S18).

Discussion

Resistance-resilient design

RRS compares each mutant score with the corresponding wild-type score after excluding wild-type magnitudes below $5.0\,\mathrm{kcal\,mol^{-1}}$. It is a docking-derived prioritization statistic, not a biochemical measurement of resistance. The classes combine docking-score magnitude with relative score retention (Table 3). Class A* combines a wild-type score magnitude of at least $7.0\,\mathrm{kcal\,mol^{-1}}$ with RRS values of at least 80 % across all tested mutants. It is a prioritization tier, not evidence of resistance circumvention. Classes B and C identify

Table 7: *Set-C MD pilot mutant-state MM-GBSA estimates (secondary, exploratory analysis; not merged into the primary docking-RRS classification). ΔG_{bind} is the mean over 100 snapshots ± SEM; the corresponding propagated within-trajectory standard deviations are reported in the Supporting Information. MMG-RRS is the mutant/WT ΔG_{bind} ratio (×100, same convention as docking RRS; WT = 100); docking RRS is shown for comparison (n/a = no WT docking reference in the pilot for PP-02 PfDHFR).*

Candidate	Target	State	ΔG_{bind} (kcal mol^{-1})	SEM	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	−35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	−29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	−30.61	1.65	100.0	–
PP-01	PfDHFR	C59R	−26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	−26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	−29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	−30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	−25.29	2.94	100.0	–
PP-02	PfCRT	K76A	−24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	−30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	−25.08	2.56	100.0	–
PP-02	PfDHFR	C59R	−28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	−31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	−34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	−27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	−30.16	2.06	100.0	–

progressively narrower resilience profiles, while Class D denotes that no available mutant reaches the 80 % retention threshold. These categories can be aligned with regional resistance surveillance in a future decision analysis, but they do not justify clinical deployment or therapeutic recommendations on their own[26].

This distinction is relevant in the current epidemiological context. The 2025 World Malaria Report documents declining efficacy of some artemisinin partner drugs alongside the spread of *Pfkelch13*-mediated partial resistance, with both trends concentrated in the African Region[1]. Resistance-aware prioritization can therefore be a useful design criterion rather than a guarantee of clinical relevance: compounds selected without an explicit mutant-sensitivity analysis may be poorly matched to the resistance states that are relevant in a given parasite population, but computational scores alone cannot establish that mismatch or its therapeutic consequence.

The relationship between RRS and molecular topology remains unresolved. The companion TDA analysis reported associations in separate overlapping cohorts[27]; because those candidates share provenance with the present study, the estimates are not independent validation. Topological descriptors should therefore be tested prospectively in larger, matched panels with mutant-state free-energy or experimental measurements before being used as a design rule.

Chemical and network context

ACSI positions candidates relative to approved-drug and African-natural-product reference spaces. The mean ACSI was 0.543, and two of 17 candidates (11.8 %) exceeded 0.70. PNS supplies a network-weighted ranking heuristic; in the target-balanced cohort, its association with RRS was weak and uncertain ($\rho = -0.2098$, adjusted $p = 1.0000$). The ACSI, PNS, and RRS measures therefore provide complementary ranking dimensions rather than independent biological measurements. Their interpretation remains conditional on the reference spaces, network construction, and target coverage[9,11,12].

Role of structural follow-up

The targeted MD analysis illustrates the role of structural follow-up. Two of four parent-study systems retained bound ligands, whereas the 164-labelled PfClpR and PfDHFR–201 systems dissociated; only PfCRT–214 passed the bound-state and topology checks for MM-GBSA. Its $(-18.25 \pm 0.40)\,\mathrm{kcal\,mol^{-1}}$ value is a system-specific endpoint estimate and cannot calibrate the Set-C candidates. Consistent with the endpoint-method literature, MM-GBSA is interpreted here as a within-protocol guide rather than converged thermodynamics[16,17]. In the Set-C pilot, mutant MM-GBSA did not reproduce the docking-predicted weakening; whether docking RRS predicts mutant free-energy resilience therefore remains unresolved.

Toward experimental validation. None of the computational estimands constitutes biochemical or biophysical evidence. The next step is matched biochemical testing of the highest-priority candidates against wild-type and mutant PfDHFR, together with orthogonal PfCRT binding or transport assays under experimentally relevant conditions. Concordant or discordant results would identify which computational layer is useful for each target class.

Comparison with existing polypharmacology tools

This framework separates docking-derived mutant sensitivity, chemical-space positioning, network-weighted ranking, and structural scrutiny. These quantities are complementary prioritization measures; none measures mutant-state binding, causal phenotype, or mechanism of action[9,27].

Limitations

The mutant structures were generated by homology modeling, introducing structural uncertainty. The available MD comprises a targeted 10-ns check of four parent-study complexes and a single-replicate 16-system Set-C pilot; neither design resolves long-timescale residence or convergence. The independent PP-01 PfCRT wild-type run differed from the canonical endpoint by 2.01 kcal mol^{-1}, treated as an empirical replicate-noise floor because snapshots are autocorrelated (Table S11). MM-GBSA is an approximate endpoint method without configurational entropy; only PfCRT–214 passed the bound-state and topology checks. The four-target panel is not a complete parasite targetome, and experimental confirmation remains necessary.

The RRS frequencies reflect the selection filters and unequal target coverage and should not be extrapolated to the full library. PfCRT network centrality was imputed, so PNS depends on the network construction (Table S7).

Ligand and protein force fields differed between cohorts, so absolute MM-GBSA values are not compared across protocols; only within-protocol contrasts are interpreted.

The target-balanced cross-metric analysis includes only $n = 12$ selected candidates, resulting in wide uncertainty intervals. Larger independent cohorts with complete target panels are needed.

PfCRT was prepared at physiological rather than digestive-vacuole pH; its docking and MD results are therefore conditional because pH correction can materially alter rankings and affinity estimates[15].

Five candidates lacked an eligible PfDHFR WT denominator and were included only in the PfCRT sensitivity analysis. This reflects the receptor and protonation setup rather than biological selectivity. Across the prespecified threshold-sensitivity grid, the baseline available-target class counts were unchanged when the WT eligibility threshold was varied from $4.0\,\mathrm{kcal\,mol^{-1}}$ to $5.0\,\mathrm{kcal\,mol^{-1}}$; tightening the retention thresholds from 70 %/80 % to 80 %/90 % increased the number of Class D candidates from 1 to 9 (Table S12).

Conclusion

This study prioritizes antimalarial candidates from African natural-product chemical space using several computational estimands. Resistance and binding claims are limited to those estimands. Among the 17 Set-C candidates, the target-balanced docking-RRS classification was A*: 1, A: 1, B: 4, C: 5, and D: 1; the coverage-sensitive available-target analysis yielded A*: 5, A: 1, B: 5, C: 5, and D: 1. The ACSI mean was 0.543. The target-balanced PNS–RRS association was weak and uncertain; the more negative 17-candidate association was coverage-sensitive and did not survive Bonferroni adjustment. These results support resistance-aware target-level ranking within the selected computational panel; they do not demonstrate biological polypharmacology or a systems-level MoA. Lightweight sensitivity analyses further show that the cross-metric associations and operational RRS classes are conditional on cohort composition and score perturbation. An independent computational replication on 39 externally sourced ligands across 8 PfDHFR/PfCRT states produced 312 finite docking records with a bootstrap mean RRS of 100.45 (99.59–101.32) and a Class-A fraction of 0.974 (0.923–101.000); this external panel used different compound sources and preparation protocols and therefore provides computational sensitivity evidence rather than experimental validation (Table S16). The four-complex MD check also showed the value of structural follow-up: two systems retained bound ligands, but only PfCRT–214 produced an interpretable MM-GBSA estimate. The study keeps potency, network context, chemical-space positioning, mutant sensitivity, and trajectory stability as separate estimands. Experimental target engagement, phenotypic MoA, mutant-state MD, and prospective validation remain necessary next steps.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T. and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. A versioned snapshot of this repository will additionally be archived on Zenodo to provide a permanent, citable DOI concurrent with publication. The large parent-study trajectories are not included in this article's data record; the conclusions drawn here are restricted to the trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

Supporting Information Available

Complete docking-validation, Tartarus calibration, MPO-sensitivity, ADMET, ACSI-sensitivity, PNS-imputation-sensitivity, and Set-C MM-GBSA tables; PlasmoDB target-identifier annotation table; MPO-sensitivity, RRS-radar, and Set-C RRS-scatter figures (PDF).

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TOC Graphic